

Course 4301

4 Credits: 0

Program Core

Source-grounded

Web Scripting Languages

- To understand the principles of creating an effective web page.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 4301 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4301.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Robotic Process Automation
Course code	4301
Course title	Web Scripting Languages
Semester	4 Credits: 0
Course category	Program Core
Periods per week	3 (L: 3 T: 0 P: 0) Periods per semester: 45
Periods per semester	45

Item	Official information
Credits	0
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

28 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To understand the principles of creating an effective web page.
- To understand the concepts of JavaScript.
- To implement client side interface through DOM, JQuery & AJAX.

Course outcomes

CO	Official outcome	Study level
CO1	web, client server architecture and formulation of web 10 Understand pages using HTML. Illustrate the concepts of JavaScript and validate web	See official syllabus
CO2	11 Understand pages using JavaScript.	See official syllabus
CO3	Apply DOM to add functionality to web pages. 10 Applying. Use Ajax, JQuery & AngularJS to enhance the	See official syllabus
CO4	12 Applying. functionality of web pages..	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 3 3
CO2	CO2 3 3 3
CO3	CO3 3 3 3
CO4	CO4 3 3 3 3 3

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	architecture and formulation of web pages using HTML.	8	As specified
Module 2	: Illustrate the concepts of JavaScript and validate web pages using JavaScript.	7	As specified
Module 3	: Apply DOM to add functionality to web pages.	5	As specified
Module 4	: Use Ajax, JQuery & Angular JS to enhance the functionality of web pages.	8	As specified

MODULE 1

architecture and formulation of web pages using HTML.

This module is presented from the official syllabus scope for Web Scripting Languages. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: architecture and formulation of web pages using HTML.
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

1 Understanding world web addresses (IP, Domain, URL), Summarize Client server architecture, web browser and

1 Understanding world web addresses (IP, Domain, URL), Summarize Client server architecture, web browser and is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding world web addresses (IP, Domain, URL), Summarize Client server architecture, web browser and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding world web addresses (ip, domain, url), summarize client server architecture, web browser and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding world web addresses (IP, Domain, URL), Summarize Client server architecture, web browser and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-1 1 Understanding world web addresses (IP, Domain, URL), Summarize Client server architecture, web browser and definition/meaning . steps, application . Technical terms English- .

1 Understanding web server communication.

1 Understanding web server communication. is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding web server communication. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding web server communication. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding web server communication. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding web server communication.
 ഹാർഡ്‌വെയർ ഹാർഡ്‌വെയർ definition/meaning ഹാർഡ്‌വെയർ ഹാർഡ്‌വെയർ, steps, application ഹാർഡ്‌വെയർ ഹാർഡ്‌വെയർ ഹാർഡ്‌വെയർ. Technical terms English- ഹാർഡ്‌വെയർ ഹാർഡ്‌വെയർ.

Define XML, Summarize XML with example. 1 Understanding Define HTML, Show basic structure of HTML,

Define XML, Summarize XML with example. 1 Understanding Define HTML, Show basic structure of HTML, is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define XML, Summarize XML with example. 1 Understanding Define HTML, Show basic structure of HTML, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define xml, summarize xml with example. 1 understanding define html, show basic structure of html, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define XML, Summarize XML with example. 1 Understanding Define HTML, Show basic structure of HTML, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 1-
Define XML, Summarize XML with example. 1
Understanding Define HTML, Show basic structure of HTML, definition/meaning . steps, application
Technical terms English-
.

Summarize head, title and metadata in HTML 1 Understanding Summarize heading tags in HTML. Summarize paragraph and line break tags in HTML,

Summarize head, title and metadata in HTML 1 Understanding Summarize heading tags in HTML. Summarize paragraph and line break tags in HTML, is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize head, title and metadata in HTML 1 Understanding Summarize heading tags in HTML. Summarize paragraph and line break tags in HTML, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize head, title and metadata in html 1 understanding summarize heading tags in html. summarize paragraph and line break tags in html, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize head, title and metadata in HTML 1 Understanding Summarize heading tags in HTML. Summarize paragraph and line break tags in HTML, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-
Summarize head, title and metadata in HTML 1
Understanding Summarize heading tags in HTML. Summarize paragraph and line break tags in HTML, definition/meaning .
 , steps, application . Technical terms English- .

Summarize Lists in HTML, Summarize Tables in 1 Understanding HTML Summarize Image tag and its attributes in HTML,

Summarize Lists in HTML, Summarize Tables in 1 Understanding HTML

Summarize Image tag and its attributes in HTML, is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize Lists in HTML, Summarize Tables in 1 Understanding HTML Summarize Image tag and its attributes in HTML, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize lists in html, summarize tables in 1 understanding html summarize image tag and its attributes in html, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize Lists in HTML, Summarize Tables in 1 Understanding HTML Summarize Image tag and its attributes in HTML, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-
1 Understand HTML Summarize Lists in HTML, Summarize Tables in HTML, Summarize Image tag and its attributes in HTML, definition/meaning . , steps, application . Technical terms English-
.

Summarize hyperlinks in HTML, Summarize Form tag 2 Understanding HTML. Summarize Form elements in HTML (Text boxes,

Summarize hyperlinks in HTML, Summarize Form tag 2 Understanding HTML. Summarize Form elements in HTML (Text boxes, is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize hyperlinks in HTML, Summarize Form tag 2 Understanding HTML. Summarize Form elements in HTML (Text boxes, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize hyperlinks in html, summarize form tag 2 understanding html. summarize form elements in html (text boxes, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize hyperlinks in HTML, Summarize Form tag 2 Understanding HTML. Summarize Form elements in HTML (Text boxes, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-
Summarize hyperlinks in HTML, Summarize Form tag 2 Understanding HTML. Summarize Form elements in HTML (Text boxes, definition/meaning . , steps, application . Technical terms English-
.

1 Understanding Radio Buttons, Check Boxes, Text Area and Buttons.) Develop a web page using all the above-mentioned

1 Understanding Radio Buttons, Check Boxes, Text Area and Buttons.) Develop a web page using all the above-mentioned is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Radio Buttons, Check Boxes, Text Area and Buttons.) Develop a web page using all the above-mentioned using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding radio buttons, check boxes, text area and buttons.) develop a web page using all the above-mentioned should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Radio Buttons, Check Boxes, Text Area and Buttons.) Develop a web page using all the above-mentioned means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പാഠ്യ 1 Understanding Radio Buttons, Check Boxes, Text Area and Buttons.) Develop a web page using all the above-mentioned ഡിഫിനിഷൻ/മനസ്സിലാക്കൽ definition/meaning പാഠ്യപരിചയം. പാഠ്യപരിചയം പാഠ്യപരിചയം, steps, application പാഠ്യപരിചയം പാഠ്യപരിചയം പാഠ്യപരിചയം. Technical terms English-ൽ പാഠ്യപരിചയം പാഠ്യപരിചയം.

2 Applying HTML elements. Addresses (IP, Domain, URL) -Client Server Architecture-Web Browser and Web Sever communication, XML. Describe HTML, Basic structure of HTML page, Basic Mark-up: -Headings - Paragraphs - line breaks -Lists - Tables-Image Tag and Attributes- Hyperlinks-Form, Form elements in HTML: - Text boxes, Radio Buttons, Check Boxes, Text Area and Buttons.

2 Applying HTML elements. Addresses (IP, Domain, URL) -Client Server Architecture-Web Browser and Web Sever communication, XML. Describe HTML, Basic structure of HTML page, Basic Mark-up: -Headings - Paragraphs - line breaks -Lists - Tables-Image Tag and Attributes- Hyperlinks-Form, Form elements in HTML: - Text boxes, Radio Buttons, Check Boxes, Text Area and Buttons. is an official topic in Module 1 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying HTML elements. Addresses (IP, Domain, URL) -Client Server Architecture-Web Browser and Web Sever communication, XML. Describe HTML, Basic structure of HTML page, Basic Mark-up: -Headings - Paragraphs - line breaks -Lists - Tables-Image Tag and Attributes- Hyperlinks-Form, Form elements in HTML: - Text boxes, Radio Buttons, Check Boxes, Text Area and Buttons. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying html elements. addresses (ip, domain, url) -client server architecture-web browser and web sever communication, xml. describe html, basic structure of html page, basic mark-up: -headings - paragraphs - line breaks -lists - tables-image tag and attributes- hyperlinks-form, form elements in html: - text boxes, radio buttons, check boxes, text area and buttons. should be discussed with attention to

safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying HTML elements. Addresses (IP, Domain, URL) -Client Server Architecture-Web Browser and Web Sever communication, XML. Describe HTML, Basic structure of HTML page, Basic Mark-up: -Headings - Paragraphs - line breaks -Lists - Tables-Image Tag and Attributes-Hyperlinks-Form, Form elements in HTML: - Text boxes, Radio Buttons, Check Boxes, Text Area and Buttons. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Applying HTML elements. Addresses (IP, Domain, URL) -Client Server Architecture-Web Browser and Web Sever communication, XML. Describe HTML, Basic structure of HTML page, Basic Mark-up: -Headings - Paragraphs - line breaks -Lists - Tables-Image Tag and Attributes-Hyperlinks-Form, Form elements in HTML: - Text boxes, Radio Buttons, Check Boxes, Text Area and Buttons. definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

architecture and formulation of web pages using HTML.		
Define Internet and world wide web, web browser and		
M1.01		1
Understanding		
world web addresses (IP, Domain, URL),		
Summarize Client server architecture, web browser and		
M1.02		1
Understanding		
web server communication.		
M1.03 Define XML, Summarize XML with example.		
Understanding		1
Define HTML, Show basic structure of HTML,		
M1.04 Summarize head, title and metadata in HTML		1
Understanding		
Summarize heading tags in HTML.		
Summarize paragraph and line break tags in HTML,		
M1.05 Summarize Lists in HTML, Summarize Tables in		1
Understanding		
HTML		
Summarize Image tag and its attributes in HTML,		
M1.06 Summarize hyperlinks in HTML, Summarize Form tag		2
Understanding		
HTML.		
Summarize Form elements in HTML (Text boxes,		
M1.07		1

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

: Illustrate the concepts of JavaScript and validate web pages using JavaScript.

This module is presented from the official syllabus scope for Web Scripting Languages. Use the topic cards for learning and the source transcript for exact

coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: : Illustrate the concepts of JavaScript and validate web pages using JavaScript.
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

webpages more interactive, examples of JavaScript in 1 Understanding browser. Basic JavaScript instructions: statements, comments,

webpages more interactive, examples of JavaScript in 1 Understanding browser. Basic JavaScript instructions: statements, comments, is an official topic in Module 2 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify webpages more interactive, examples of JavaScript in 1 Understanding browser. Basic JavaScript instructions: statements, comments, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, webpages more interactive, examples of javascript in 1 understanding browser. basic javascript instructions: statements, comments, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What webpages more interactive, examples of JavaScript in 1 Understanding browser. Basic JavaScript instructions: statements, comments, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[webpages](#) more interactive, examples of JavaScript in 1 Understanding browser. Basic JavaScript instructions: statements, comments, [definition/meaning](#). [steps](#), application [Technical terms](#) English-.

1 Understanding variable, data types.

1 Understanding variable, data types. is an official topic in Module 2 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding variable, data types. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding variable, data types. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding variable, data types. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-1 Understanding variable, data types.

പ്രാഥമികമായി definition/meaning ഉൾപ്പെടെ. ഉൾപ്പെടെ definition, steps, application ഉൾപ്പെടെ ഉൾപ്പെടെ. Technical terms English-ൽ ഉൾപ്പെടെ ഉൾപ്പെടെ.

JavaScript arrays, expressions, operators. 1 Understanding Functions methods: function, anonymous function,

JavaScript arrays, expressions, operators. 1 Understanding Functions methods: function, anonymous function, is an official topic in Module 2 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify JavaScript arrays, expressions, operators. 1 Understanding Functions methods: function, anonymous function, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, javascript arrays, expressions, operators. 1 understanding functions methods: function, anonymous function, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What JavaScript arrays, expressions, operators. 1 Understanding Functions methods: function, anonymous function, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
JavaScript arrays, expressions, operators. 1
Understanding Functions methods: function, anonymous function,
definition/meaning . . . , steps,
application Technical terms English-
.

2 Understanding variable scope. Functions objects: object, this, arrays are objects,

2 Understanding variable scope. Functions objects: object, this, arrays are objects, is an official topic in Module 2 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding variable scope. Functions objects: object, this, arrays are objects, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding variable scope. functions objects: object, this, arrays are objects, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding variable scope. Functions objects: object, this, arrays are objects, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഘട്ടം 2 Understanding variable scope. Functions objects: object, this, arrays are objects, നിർവ്വചന/അർത്ഥം definition/meaning നിർവ്വചനം. ഘട്ടങ്ങൾ ഘട്ടങ്ങൾ, steps, application പ്രയോഗം നിർവ്വചനം. Technical terms English-ഘട്ടം നിർവ്വചനം നിർവ്വചനം.

2 Understanding browser object model, document object model.

2 Understanding browser object model, document object model. is an official topic in Module 2 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding browser object model, document object model. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding browser object model, document object model. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding browser object model, document object model. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2- 2 Understanding browser object model, document object model. ഫലപ്രദമായ രീതിയിൽ definition/meaning നൽകുക. ഫലപ്രദമായ രീതിയിൽ steps, application നൽകുക. Technical terms English-ൽ നൽകുക.

Global objects: string, number, math, date. 1 Understanding Decision making and Loops: decision making: if statement, if...else statement, switch statement, loops:

Global objects: string, number, math, date. 1 Understanding Decision making and Loops: decision making: if statement, if...else statement, switch statement, loops: is an official topic in Module 2 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Global objects: string, number, math, date. 1 Understanding Decision making and Loops: decision making: if statement, if...else statement, switch statement, loops: using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, global objects: string, number, math, date. 1 understanding decision making and loops: decision making: if statement, if...else statement, switch statement, loops: should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Global objects: string, number, math, date. 1 Understanding Decision making and Loops: decision making: if statement, if...else statement, switch statement, loops: means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-Global objects: string, number, math, date. 1 Understanding Decision making and Loops: decision making: if statement, if...else statement, switch statement, loops: definition/meaning, steps, application. Technical terms English-

3 Applying key loop concepts, for loops, while loops, do while loops. Define JavaScript – How JavaScript makes the webpages more interactive - examples of JavaScript in browser, Basic JavaScript instructions: statements, comments, variable, data types, arrays, expressions, operators. Functions methods: function, anonymous function, variable scope, object, this, arrays are objects, browser object model, document object model, Global objects: string, number, math, date. Decision making and Loops: decision making: if statement, if...else statement, switch statement, loops: key loop concepts, for loops, while loops, do while loops.

3 Applying key loop concepts, for loops, while loops, do while loops. Define JavaScript – How JavaScript makes the webpages more interactive - examples of JavaScript in browser, Basic JavaScript instructions: statements, comments, variable, data types, arrays, expressions, operators. Functions methods: function, anonymous function, variable scope, object, this, arrays are objects, browser object model, document object model, Global objects: string, number, math, date. Decision making and Loops: decision making: if statement, if...else statement, switch statement, loops: key loop concepts, for loops, while loops, do while loops. is an official topic in Module 2 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying key loop concepts, for loops, while loops, do while loops. Define JavaScript – How JavaScript makes the webpages more interactive - examples of JavaScript in browser, Basic JavaScript instructions: statements, comments, variable, data types, arrays, expressions, operators. Functions methods: function, anonymous function, variable scope, object, this, arrays are objects, browser object model, document object model, Global objects: string, number, math, date. Decision making and Loops: decision making: if statement, if...else statement, switch statement, loops: key loop concepts, for loops, while loops, do while loops. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying key loop concepts, for loops, while loops, do while loops. define javascript – how javascript makes the webpages more interactive - examples of javascript in browser, basic javascript instructions: statements, comments, variable, data types, arrays, expressions, operators. functions methods: function, anonymous function, variable scope, object, this, arrays are objects, browser object model, document object model, global objects: string, number, math, date. decision making and loops: decision making: if statement, if...else statement, switch statement, loops: key loop concepts, for loops, while loops, do while loops. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying key loop concepts, for loops, while loops, do while loops. Define JavaScript – How JavaScript makes the webpages more interactive - examples of JavaScript in browser, Basic JavaScript instructions: statements, comments, variable, data types, arrays, expressions, operators. Functions methods: function, anonymous function, variable scope, object, this, arrays are objects, browser object model, document object model, Global objects: string, number, math, date. Decision making and Loops: decision making: if statement, if...else statement, switch statement, loops: key loop concepts, for loops, while loops, do while loops. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 3 Applying key loop concepts, for loops, while loops, do while loops. Define JavaScript – How JavaScript makes the webpages more interactive - examples of JavaScript in browser, Basic JavaScript instructions: statements, comments, variable, data types, arrays, expressions, operators. Functions methods: function, anonymous function, variable scope, object, this, arrays are objects, browser object model, document object model, Global objects: string, number, math, date. Decision making and Loops: decision making: if statement, if...else statement, switch statement, loops: key loop concepts, for

loops, while loops, do while loops. ലൂപ്പ്, വൈൽ ലൂപ്പ്, ഡോ വൈൽ ലൂപ്പ് definition/meaning അർത്ഥം/അർത്ഥം. ലൂപ്പ് വൈൽ ലൂപ്പ് ഡോ വൈൽ ലൂപ്പ്, steps, application പ്രയോഗം പ്രയോഗം പ്രയോഗം. Technical terms English-മലയാളം മലയാളം മലയാളം.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

: Illustrate the concepts of JavaScript and validate web pages using JavaScript. Define JavaScript, How JavaScript makes the		
M2.01	webpages more interactive, examples of JavaScript in browser.	1
Understanding		
	Basic JavaScript instructions: statements, comments,	
M2.02		1
Understanding	variable, data types.	
	M2.03 JavaScript arrays, expressions, operators.	1
Understanding		
	Functions methods: function, anonymous function,	
M2.04		2
Understanding	variable scope.	
	Functions objects: object, this, arrays are objects,	
M2.05		2
Understanding	browser object model, document object model.	
	M2.06 Global objects: string, number, math, date.	1
Understanding		
	Decision making and Loops: decision making: if statement, if...else statement, switch statement, loops:	

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

: Apply DOM to add functionality to web pages.

This module is presented from the official syllabus scope for Web Scripting Languages. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: : Apply DOM to add functionality to web pages.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

2 Understanding (DOM), DOM tree as a model of a web page. DOM tree - working with DOM tree accessing

2 Understanding (DOM), DOM tree as a model of a web page. DOM tree - working with DOM tree accessing is an official topic in Module 3 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding (DOM), DOM tree as a model of a web page. DOM tree - working with DOM tree accessing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding (dom), dom tree as a model of a web page. dom tree - working with dom tree accessing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding (DOM), DOM tree as a model of a web page. DOM tree - working with DOM tree accessing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-2 Understanding (DOM), DOM tree as a model of a web page. DOM tree - working with DOM tree accessing definition/meaning definition/meaning steps, application Technical terms English-

elements, nodelists, selecting elements: using class 2 Understanding attribute, tag name. DOM - Traversing, adding or removing html content,

elements, nodelists, selecting elements: using class 2 Understanding attribute, tag name. DOM - Traversing, adding or removing html content, is an official topic in Module 3 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify elements, nodelists, selecting elements: using class 2 Understanding attribute, tag name. DOM - Traversing, adding or removing html content, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, elements, nodelists, selecting elements: using class 2 understanding attribute, tag name. dom - traversing, adding or removing html content, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What elements, nodelists, selecting elements: using class 2 Understanding attribute, tag name. DOM - Traversing, adding or removing html content, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- π elements, nodelists, selecting elements: using class 2 Understanding attribute, tag name. DOM - Traversing, adding or removing html content, π π definition/meaning π . π π π , steps, application π π π . Technical terms English- π π π .

2 Applying. update text and markup, adding/removing elements. Event handling: different event types, three ways to

2 Applying. update text and markup, adding/removing elements. Event handling: different event types, three ways to is an official topic in Module 3 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying. update text and markup, adding/removing elements. Event handling: different event types, three ways to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying. update text and markup, adding/removing elements. event handling: different event types, three ways to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying. update text and markup, adding/removing elements. Event handling: different event types, three ways to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓ 2 Applying. update text and markup, adding/removing elements. Event handling: different event types, three ways to ഓരോന്നിനും ഓരോന്നും definition/meaning ഓരോന്നിനും. ഓരോന്നും ഓരോന്നും ഓരോന്നും, steps, application ഓരോന്നും ഓരോന്നിനും ഓരോന്നും. Technical terms English-ഓ ഓരോന്നും ഓരോന്നിനും.

bind an event to an element, using DOM event 2 Applying. handlers. Describe event listeners, using parameters with event

bind an event to an element, using DOM event 2 Applying. handlers. Describe event listeners, using parameters with event is an official topic in Module 3 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify bind an event to an element, using DOM event 2 Applying. handlers. Describe event listeners, using parameters with event using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, bind an event to an element, using dom event 2 applying. handlers. describe event listeners, using parameters with event should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What bind an event to an element, using DOM event 2 Applying. handlers. Describe event listeners, using parameters with event means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എന്നത് bind an event to an element, using DOM event 2 Applying. handlers. Describe event listeners, using parameters with event definition/meaning. steps, application Technical terms English-എന്നത്

listeners, the event object, event delegation, user 2 Applying. interface events, event bubbling. a web page, working with DOM tree, accessing elements, nodelists, selecting elements: using class attribute, tag name. DOM - Traversing, adding or removing html content, update text and markup, adding/removing elements. Event handling: different event types, three ways to bind an event to an element, using DOM event handlers. Describe event listeners, using parameters with event listeners, the event object, event delegation, user interface events, event bubbling.

listeners, the event object, event delegation, user 2 Applying. interface events, event bubbling. a web page, working with DOM tree, accessing elements, nodelists, selecting elements: using class attribute, tag name. DOM - Traversing, adding or removing html content, update text and markup, adding/removing elements. Event handling: different event types, three ways to bind an event to an element, using DOM event handlers. Describe event listeners, using parameters with event listeners, the event object, event delegation, user interface events, event bubbling. is an official topic in Module 3 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify listeners, the event object, event delegation, user 2 Applying. interface events, event bubbling. a web page, working with DOM tree, accessing elements, nodelists, selecting elements: using class attribute, tag name. DOM - Traversing, adding or removing html content, update text and markup, adding/removing elements. Event handling: different event types, three ways to bind an event to an element, using DOM event handlers. Describe event listeners, using parameters with event listeners, the event object, event delegation, user interface events, event bubbling. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, listeners, the event object, event delegation, user interface events, event bubbling. a web page, working with dom tree, accessing elements, nodelists, selecting elements: using class attribute, tag name. dom - traversing, adding or removing html content, update text and markup, adding/removing elements. event handling: different event types, three ways to bind an event to an element, using dom event handlers. describe event listeners, using parameters with event listeners, the event object, event delegation, user interface events, event bubbling. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What listeners, the event object, event delegation, user 2 Applying. interface events, event bubbling. a web page, working with DOM tree, accessing elements, nodelists, selecting elements: using class attribute, tag name. DOM - Traversing, adding or removing html content, update text and markup, adding/removing elements. Event handling: different event types, three ways to bind an event to an element, using DOM event handlers. Describe event listeners, using parameters with event listeners, the event object, event delegation, user interface events, event bubbling. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
listeners, the event object, event delegation, user 2 Applying. interface events, event bubbling. a web page, working with DOM tree, accessing elements, nodelists, selecting elements: using class attribute, tag name. DOM - Traversing, adding or removing html content, update text and markup, adding/removing elements. Event handling: different event types, three ways to bind an event to an element, using DOM event handlers. Describe event listeners, using parameters with event listeners, the event object, event delegation, user interface events, event bubbling. definition/meaning . steps, application . Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

```
: Apply DOM to add functionality to web pages.
DOM:Introduction to Document Object Model
M3.01 Understanding (DOM), DOM tree as a model of a web page. 2
DOM tree - working with DOM tree accessing
M3.02 elements, nodelists, selecting elements: using class 2
Understanding attribute, tag name.
DOM - Traversing, adding or removing html content,
M3.03 Applying. 2
update text and markup, adding/removing elements.
Event handling: different event types, three ways to
M3.04 bind an event to an element, using DOM event 2
Applying. handlers.
Describe event listeners, using parameters with event
M3.05 listeners, the event object, event delegation, user 2
Applying. interface events, event bubbling.
Contents: DOM: Introduction to Document Object Model (DOM), DOM tree as a model
of
a web page, working with DOM tree, accessing elements, nodelists, selecting
elements:
using class attribute, tag name. DOM - Traversing, adding or removing html
content, update
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

: Use Ajax, JQuery & Angular JS to enhance the functionality of web pages.

This module is presented from the official syllabus scope for Web Scripting Languages. Use the topic cards for learning and the source transcript for exact

coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: : Use Ajax, JQuery & Angular JS to enhance the functionality of web pages.
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

1 Understanding works? Handling Ajax request and response, data format:

1 Understanding works? Handling Ajax request and response, data format: is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding works? Handling Ajax request and response, data format: using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding works? handling ajax request and response, data format: should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding works? Handling Ajax request and response, data format: means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-പാഠ്യ 1 Understanding works? Handling Ajax request and response, data format: പരസ്പരം പരസ്പരം definition/meaning പരസ്പരം പരസ്പരം. പരസ്പരം പരസ്പരം പരസ്പരം, steps, application പരസ്പരം പരസ്പരം പരസ്പരം. Technical terms English-പാ പരസ്പരം പരസ്പരം പരസ്പരം.

1 Applying. XML. JSON – Introduction to JSON, Working with JSON

1 Applying. XML. JSON – Introduction to JSON, Working with JSON is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Applying. XML. JSON – Introduction to JSON, Working with JSON using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 applying. xml. json – introduction to json, working with json should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Applying. XML. JSON – Introduction to JSON, Working with JSON means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4- 1 Applying. XML. JSON – Introduction to JSON, Working with JSON ഓരോന്നിന്റെയും definition/meaning ഓരോന്നിന്റെയും. ഓരോന്നിന്റെയും steps, application ഓരോന്നിന്റെയും. Technical terms English-ൽ ഓരോന്നിന്റെയും.

2 Applying data, Loading HTML & JSON with Ajax. JQUERY : What is JQuery ?, A basic JQuery

2 Applying data, Loading HTML & JSON with Ajax. JQUERY : What is JQuery ?, A basic JQuery is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying data, Loading HTML & JSON with Ajax. JQUERY : What is JQuery ?, A basic JQuery using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying data, loading html & json with ajax. jquery : what is jquery ?, a basic jquery should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying data, Loading HTML & JSON with Ajax. JQUERY : What is JQuery ?, A basic JQuery means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-2 Applying data, Loading HTML & JSON with Ajax. JQUERY : What is JQuery ?, A basic JQuery definition/meaning, steps, application Technical terms English-

2 Understanding example, Why use JQuery ? JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements,

2 Understanding example, Why use JQuery ? JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding example, Why use JQuery ? JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding example, why use jquery ? jquery operations - finding elements, jquery selection, getting element content, updating elements, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding example, Why use JQuery ? JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പാഠ്യ 2 Understanding example, Why use JQuery ? JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, പദപരിവർത്തനം പദപരിവർത്തനം definition/meaning പദപരിവർത്തനം. പദപരിവർത്തനം പദപരിവർത്തനം, steps, application പദപരിവർത്തനം പദപരിവർത്തനം പദപരിവർത്തനം. Technical terms English-ൽ പദപരിവർത്തനം പദപരിവർത്തനം.

2 Applying changing content, inserting elements, adding new content, getting and setting attributes. Angular JS: What is Angular JS?, Why use Angular

2 Applying changing content, inserting elements, adding new content, getting and setting attributes. Angular JS: What is Angular JS?, Why use Angular is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying changing content, inserting elements, adding new content, getting and setting attributes. Angular JS: What is Angular JS?, Why use Angular using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying changing content, inserting elements, adding new content, getting and setting attributes. angular js: what is angular js?, why use angular should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying changing content, inserting elements, adding new content, getting and setting attributes. Angular JS: What is Angular JS?, Why use Angular means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 2 Applying changing content, inserting elements, adding new content, getting and setting attributes. Angular JS: What is Angular JS?, Why use Angular **പ്രശ്നം** definition/meaning **പ്രശ്നം**. **പ്രശ്നം** **പ്രശ്നം** **പ്രശ്നം**, steps, application **പ്രശ്നം** **പ്രശ്നം** **പ്രശ്നം**. Technical terms English- **പ്രശ്നം** **പ്രശ്നം**.

1 Understanding JS?, How Angular JS works?.

1 Understanding JS?, How Angular JS works?.. is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding JS?, How Angular JS works?.. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding js?, how angular js works?.. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding JS?, How Angular JS works?.. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-1 Understanding JS?, How Angular JS works?.
 ഓരോന്നിനും definition/meaning ഉൾപ്പെടെ. ഓരോന്നിനും steps, application ഉൾപ്പെടെ. Technical terms English-ൽ ഉൾപ്പെടെ.
 ഉൾപ്പെടെ.

Angular Expression, Filters, Directives, Controllers.

Angular Expression, Filters, Directives, Controllers. is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Angular Expression, Filters, Directives, Controllers. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, angular expression, filters, directives, controllers. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Angular Expression, Filters, Directives, Controllers. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Angular Expression, Filters, Directives, Controllers. definition/meaning .
 , steps, application . Technical terms English- .

Familiarization of Angular JS modules & Forms. 2 Applying Introduction to AJAX, Handling Ajax request and response, data format: XML, Introduction to JSON, Working with JSON data, Loading HTML & JSON with Ajax, Introduction to JQUERY, JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, changing content, inserting elements, adding new content, getting and setting attributes. Introduction to Angular JS, Angular Expression, Filters, Directives, Controllers. Angular JS modules & Forms. (no detailed study required.) Text / Reference T/R Book Title/Author T1 Paul J. Deitel, Harvey M. Deitel, Abbey Deitel, "Internet and World Wide Web How To Program", 5/E, Pearson Education, 2012. R1 Jon Duckett, "HTML and CSS: Design and Build Websites", Wiley R2 Jon Duckett, "JavaScript and JQuery : Interactive Front-End Web Development", Wiley. R3 Felipe Coury, Ari Lerner, Carlos Taborda "ng - book: The Complete Guide to Angular Paperback - 6", February 2018. Online Resources Sl.No Website Link 1 <https://www.w3schools.com/html/default.asp> 2 <https://www.tutorialspoint.com/html5/index.htm> 3 <https://www.w3schools.com/js/default.asp> 4 <https://www.w3schools.com/angular>

Familiarization of Angular JS modules & Forms. 2 Applying Introduction to AJAX, Handling Ajax request and response, data format: XML, Introduction to JSON, Working with JSON data, Loading HTML & JSON with Ajax, Introduction to JQUERY, JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, changing content, inserting elements, adding new content, getting and setting attributes. Introduction to Angular JS, Angular Expression, Filters, Directives, Controllers. Angular JS modules & Forms. (no detailed study required.) Text / Reference T/R Book Title/Author T1 Paul J. Deitel, Harvey M. Deitel, Abbey Deitel, "Internet and World Wide Web How To Program", 5/E, Pearson Education, 2012. R1 Jon Duckett, "HTML and CSS: Design and Build Websites", Wiley R2 Jon Duckett, "JavaScript and JQuery : Interactive Front-End Web Development", Wiley. R3 Felipe Coury, Ari Lerner, Carlos Taborda "ng - book: The Complete Guide to Angular Paperback - 6", February 2018. Online Resources Sl.No Website Link 1 <https://www.w3schools.com/html/default.asp> 2 <https://www.tutorialspoint.com/html5/index.htm> 3 <https://www.w3schools.com/js/default.asp> 4 <https://www.w3schools.com/angular> is an official topic in Module 4 of Web Scripting Languages. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Familiarization of Angular JS modules & Forms. 2 Applying Introduction to AJAX, Handling Ajax request and response, data format: XML, Introduction to JSON, Working with JSON data, Loading HTML & JSON with Ajax, Introduction to JQUERY, JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, changing content, inserting elements, adding new content, getting and setting attributes. Introduction to Angular JS, Angular Expression, Filters, Directives, Controllers. Angular JS modules & Forms. (no detailed study required.) Text / Reference T/R Book Title/Author T1 Paul J. Deitel, Harvey M. Deitel, Abbey Deitel, "Internet and World Wide Web How To Program", 5/E, Pearson Education, 2012. R1 Jon Duckett, "HTML and CSS: Design and Build Websites", Wiley R2 Jon Duckett, "JavaScript and JQuery : Interactive Front-End Web Development", Wiley. R3 Felipe Coury, Ari Lerner, Carlos Taborda "ng - book: The Complete Guide to Angular Paperback - 6", February 2018. Online Resources Sl.No Website Link 1 <https://www.w3schools.com/html/default.asp> 2 <https://www.tutorialspoint.com/html5/index.htm> 3 <https://www.w3schools.com/js/default.asp> 4 <https://www.w3schools.com/angular> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, familiarization of angular js modules & forms. 2 applying introduction to ajax, handling ajax request and response, data format: xml, introduction to json, working with json data, loading html & json with ajax, introduction to jquery, jquery operations - finding elements, jquery selection, getting element content, updating elements, changing content, inserting elements, adding new content, getting and setting attributes. introduction to angular js, angular expression, filters, directives, controllers. angular js modules & forms. (no detailed study required.) text / reference t/r book title/author t1 paul j. deitel, harvey m. deitel, abbey deitel, "internet and world wide web how to program", 5/e, pearson education, 2012. r1 jon duckett, "html and css: design and build websites", wiley r2 jon duckett, "javascript and jquery : interactive front-end web development", wiley. r3 felipe coury, ari lerner, carlos taborda "ng - book: the complete guide to angular paperback - 6", february 2018. online

resources sl.no website link 1 <https://www.w3schools.com/html/default.asp> 2 <https://www.tutorialspoint.com/html5/index.htm> 3 <https://www.w3schools.com/js/default.asp> 4 <https://www.w3schools.com/angular> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Familiarization of Angular JS modules & Forms. 2 Applying Introduction to AJAX, Handling Ajax request and response, data format: XML, Introduction to JSON, Working with JSON data, Loading HTML & JSON with Ajax, Introduction to JQUERY, JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, changing content, inserting elements, adding new content, getting and setting attributes. Introduction to Angular JS, Angular Expression, Filters, Directives, Controllers. Angular JS modules & Forms. (no detailed study required.) Text / Reference T/R Book Title/Author T1 Paul J. Deitel, Harvey M. Deitel, Abbey Deitel, "Internet and World Wide Web How To Program", 5/E, Pearson Education, 2012. R1 Jon Duckett, "HTML and CSS: Design and Build Websites", Wiley R2 Jon Duckett, "JavaScript and JQuery : Interactive Front-End Web Development", Wiley. R3 Felipe Coury, Ari Lerner, Carlos Taborda "ng - book: The Complete Guide to Angular Paperback - 6", February 2018. Online Resources Sl.No Website Link 1 https://www.w3schools.com/html/default.asp 2 https://www.tutorialspoint.com/html5/index.htm 3 https://www.w3schools.com/js/default.asp 4 https://www.w3schools.com/angular means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Familiarization of Angular JS modules & Forms. 2 Applying Introduction to AJAX, Handling Ajax request and response, data format: XML, Introduction to JSON, Working with JSON data, Loading HTML & JSON with Ajax, Introduction to JQUERY, JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements, changing content, inserting elements, adding new content, getting and setting attributes. Introduction to Angular JS, Angular Expression, Filters, Directives, Controllers. Angular JS modules & Forms. (no detailed study required.) Text / Reference T/R Book Title/Author T1 Paul J. Deitel, Harvey M. Deitel, Abbey Deitel, "Internet and World Wide Web How To Program", 5/E, Pearson Education, 2012. R1 Jon Duckett, "HTML and CSS: Design and Build Websites", Wiley R2 Jon Duckett, "JavaScript and JQuery : Interactive Front-End Web Development", Wiley. R3 Felipe Coury, Ari Lerner, Carlos Taborda "ng - book: The Complete Guide to Angular Paperback - 6", February 2018. Online

Resources Sl.No Website Link 1 <https://www.w3schools.com/html/default.asp> 2 <https://www.tutorialspoint.com/html5/index.htm> 3 <https://www.w3schools.com/js/default.asp> 4 <https://www.w3schools.com/angular>

പദപരിഭാഷ, പദപരിഭാഷ definition/meaning പദപരിഭാഷ. പദപരിഭാഷ പദപരിഭാഷ പദപരിഭാഷ, steps, application പദപരിഭാഷ പദപരിഭാഷ പദപരിഭാഷ. Technical terms English-പദപരിഭാഷ പദപരിഭാഷ.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

: Use Ajax, JQuery & Angular JS to enhance the functionality of web pages.		
AJAX: What is Ajax?, Why use Ajax?, How Ajax		
M4.01		1
Understanding		
	works?	
	Handling Ajax request and response, data format:	
M4.02		1
Applying.		
	XML.	
	JSON – Introduction to JSON, Working with JSON	
M4.03		2
Applying		
	data, Loading HTML & JSON with Ajax.	
	JQUERY : What is JQuery ?, A basic JQuery	
M4.04		2
Understanding		
	example, Why use JQuery ?	
	JQUERY Operations - finding elements, JQuery	
	selection, getting element content, updating elements,	
M4.05		2
Applying		
	changing content, inserting elements, adding new	
	content, getting and setting attributes.	
	Angular JS: What is Angular JS?, Why use Angular	
M4.06		1
Understanding		
	JS?, How Angular JS works?.	

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 1 Understanding world web addresses (IP, Domain, URL), Summarize Client server architecture, web browser and. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding world web addresses (IP, Domain, URL), Summarize Client server architecture, web browser and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 1 Understanding web server communication.. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding web server communication..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Define XML, Summarize XML with example. 1 Understanding Define HTML,Show basic structure of HTML,. (3 marks)

Answer: Begin with the standard definition or identity of *Define XML, Summarize XML with example. 1 Understanding Define HTML,Show basic structure of HTML,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Summarize head, title and metadata in HTML 1 Understanding Summarize heading tags in HTML. Summarize paragraph and line break tags in HTML,. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize head, title and metadata in HTML* 1 Understanding *Summarize heading tags in HTML*. *Summarize paragraph and line break tags in HTML*,. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain webpages more interactive, examples of JavaScript in 1 Understanding browser. Basic JavaScript instructions: statements, comments, (3 marks)

Answer: Begin with the standard definition or identity of *webpages more interactive, examples of JavaScript in 1 Understanding browser*. *Basic JavaScript instructions: statements, comments*,. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 1 Understanding variable, data types.. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding variable, data types*.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain JavaScript arrays, expressions, operators. 1 Understanding Functions methods: function, anonymous function, (3 marks)

Answer: Begin with the standard definition or identity of *JavaScript arrays, expressions, operators. 1 Understanding Functions methods: function, anonymous function,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Understanding variable scope. Functions objects: object, this, arrays are objects,. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding variable scope. Functions objects: object, this, arrays are objects,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain 2 Understanding (DOM), DOM tree as a model of a web page. DOM tree - working with DOM tree accessing. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding (DOM), DOM tree as a model of a web page. DOM tree - working with DOM tree accessing.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain elements, nodelists, selecting elements: using class 2 Understanding attribute, tag name. DOM - Traversing, adding or removing

html content,. (3 marks)

Answer: Begin with the standard definition or identity of *elements, nodelists, selecting elements: using class 2 Understanding attribute, tag name. DOM - Traversing, adding or removing html content,.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Applying. update text and markup, adding/removing elements. Event handling: different event types, three ways to. (3 marks)

Answer: Begin with the standard definition or identity of *2 Applying. update text and markup, adding/removing elements. Event handling: different event types, three ways to.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain bind an event to an element, using DOM event 2 Applying. handlers. Describe event listeners, using parameters with event. (3 marks)

Answer: Begin with the standard definition or identity of *bind an event to an element, using DOM event 2 Applying. handlers. Describe event listeners, using parameters with event.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain 1 Understanding works? Handling Ajax request and response, data format:. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding works? Handling Ajax request and response, data format:.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 1 Applying. XML. JSON - Introduction to JSON, Working with JSON. (3 marks)

Answer: Begin with the standard definition or identity of *1 Applying. XML. JSON - Introduction to JSON, Working with JSON.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Applying data, Loading HTML & JSON with Ajax. JQUERY : What is JQuery ?, A basic JQuery. (3 marks)

Answer: Begin with the standard definition or identity of *2 Applying data, Loading HTML & JSON with Ajax. JQUERY : What is JQuery ?, A basic JQuery.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Understanding example, Why use JQuery ? JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements,. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding example, Why use JQuery ? JQUERY Operations - finding elements, JQuery selection, getting element content, updating elements*,. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **1 Understanding world web addresses (IP, Domain, URL), Summarize Client server architecture, web browser and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **webpages more interactive, examples of JavaScript in 1 Understanding browser. Basic JavaScript instructions: statements, comments,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding (DOM), DOM tree as a model of a web page. DOM tree - working with DOM tree accessing.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **1 Understanding works? Handling Ajax request and response, data format:.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.

- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- ", Wiley
- R2 Jon Duckett , "JavaScript and JQuery : Interactive Front-End Web Development", Wiley.
- R3 Felipe Coury, Ari Lerner, Carlos Taborda"ng - book: The Complete Guide to Angular Paperback - 6", February 2018. Online Resources
- Sl.No Website Link <https://www.w3schools.com/html/default.asp>
- <https://www.tutorialspoint.com/html5/index.htm>
- <https://www.w3schools.com/js/default.asp> <https://www.w3schools.com/angular>

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Prepared from the supplied official SITTR syllabus PDF for Course 4301. Verify institutional updates against the current official curriculum.